

FARHAN KERIWALA

MSc Computational Finance student and First-Class Computer Science graduate interested in Quantitative Finance

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EDUCATION

Master of Science in Computational Finance - King's College London (Sep 2025 – Sep 2026)

- Concentrations: MSc Individual Project, Quantitative Methods in Finance, Agent-Based Modelling in Finance, High-Frequency Finance, C++ for Financial Mathematics and Statistics in Finance.

Bachelor of Science with Honours in Computer Science - Queen Mary University of London (Sep 2022 – Jun 2025)

- Grade: First Class
- Concentrations: Probability Theory, Security, Formal Languages, Web & Object-Oriented Programming and Machine Learning.
- Projects: BSc Final Year Project, Django Web App, Java Swing Game, Portfolio, COOL Compiler and CIFAR Classification.

GCE Advanced Level - Hazelwick School Sixth Form (Sep 2020 – Jun 2022)

- Grades: A*, A, A, A
- Subjects: Mathematics, Further Mathematics, Computer Science and Physics.

SKILLS

- Programming & Tools: Python, Java, C++, NumPy, Pandas, Matplotlib, PyTorch, SQL, Git, Docker, Supabase and Next.js.
- Quantitative Finance & Mathematics: Black-Scholes closed-form, Monte Carlo, Greeks calculation, Stochastic calculus (Geometric Brownian Motion, Ito's Lemma), Time-series analysis and Regression models.
- Machine Learning & Data Science: Feature engineering, Sentiment analysis, Natural Language Processing, Neural networks (CNNs) and Data processing pipelines.

PROJECTS

Market Options Lab (Sep 2025 – Present)

- An options pricing and risk analysis lab that utilises Black-Scholes and Monte Carlo methods under the Geometric Brownian Motion model, including Greeks and implied volatility smiles. Validate the models with a delta-hedging backtest on real market data and present results in a modern dashboard.

Final Year BSc Research Project (Aug 2024 – May 2025)

- Predicted stock prices using over 20 years of financial and sentiment data, employing advanced feature engineering and techniques. Compared regression models to assess sentiment data's impact on predictive accuracy, achieving a 95% baseline accuracy and 84% enhanced accuracy with sentiment data.

CIFAR10 Classification Model (Mar 2025 – Apr 2025)

- A Convolutional Neural Network to classify images from the CIFAR-10 dataset into ten distinct classes, following a specified model architecture.
- Loss reduction and model optimisation techniques were applied to improve the model's performance. Ultimately, the finalised model achieved a validation accuracy of 86% and an overall project score of 95%.

Vibe – AI Application Builder (Sep 2025)

- An AI-driven full-stack application builder that converts simple prompts into deployable projects using programmable agents.
- Integrates sandboxed execution, background jobs, and a complete developer workflow, including authentication and billing.

EXPERIENCE

Sales Associate - Luxottica (Jul 2024 – Oct 2024, Jun 2025 - Present)

- Consistently exceeding sales targets; developing strong client-facing communication skills.

Lab Demonstrator - Queen Mary University of London (Sep 2024 – Apr 2025)

- Guided over 100 students and supported them through various lab tasks and coursework projects.

Virtual Developer Experience - Accenture (Jan 2024 – Apr 2024)

- Completed the Developer and Technology Job Simulation, gaining a comprehensive understanding of SDLCs, and delivered a compelling presentation that analysed and compared software methodologies.

Virtual Internship - Goldman Sachs (Dec 2023)

- Identified outdated password hashing through Hashcat, then recommended increasing the minimum password length to 12 and adopting a dedicated hashing algorithm in a memo to my supervisor.